

Kunal Chandra

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[LinkedIn](#)

[LeetCode](#)

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[Portfolio](#)

Work Experience

Defence Research and Development Organisation (DRDO), Delhi

Institute of Nuclear Medicine and Allied Sciences (INMAS)

1 Jun 2026 – 30 Nov 2026

Software Development Engineer (SDE) Intern | Ongoing

- Developing an **OCR-based visitor gate pass system** for **real-time identity verification** and automated data extraction
- Optimizing document processing pipelines using **GPU-accelerated OCR** and advanced **image preprocessing** techniques
- Building an **LLM-powered chatbot** for intelligent information retrieval and automated query resolution
- Designing **full-stack applications** with **secure database integration** and workflow automation

Solid State Physics Laboratory (SSPL)

1 Jan 2026 – 31 Jan 2026

Machine Learning Intern | [Link](#)

- Developed a **Siamese Deep Learning Architecture (SNUNet)** for satellite imagery change detection
- Enabled **defence-grade intelligence workflows** for border surveillance and infrastructure monitoring
- Achieved **96.37% Precision, 95.95% Recall and 96.16% F1 Score** on the CDD dataset

Intelligence Fusion & Strategic Operations (IFSO) Special Cell, Delhi Police

Aug 2025

Data Science Intern | [Link](#)

- Built **AI-driven Object Mapping and Dynamic Script Modification** tool for Python GUI scripts
- Detected widgets and auto-injected event-handling methods into existing code
- Used **AST parsing, regex, JSON handling** and dynamic code generation

Technical Skills

Languages: Python Java JavaScript Go C HTML CSS

Frameworks and Libraries: Pandas NumPy Matplotlib Seaborn OpenCV NLTK Scikit-learn TensorFlow Keras PyTorch YOLOv8 Flask MediaPipe FastAPI

Databases and Developer Tools: MongoDB PostgreSQL MySQL SQL GitHub Git Docker

Generative AI & LLM: LangChain LangGraph LlamaIndex Hugging Face Transformers RAG (Retrieval-Augmented Generation) Vector Databases

Projects

[Intelligent Traffic Signal Control System \(SMARTFLOW\)](#)

[Live](#)

- Developed an **AI-driven traffic management system** using **YOLOv8** for real-time vehicle detection
- Implemented robust object tracking using **BYTETrack** algorithm with real-time dashboard
- Achieved up to **30% congestion reduction** using Reinforcement Learning and LSTM-based signal timing
- Integrated **ESP32 with RFID** to prioritize emergency vehicles

[Pravha AI Powered Flood Forecasting and Emergency Response System](#)

[Live](#)

- Built proactive flood forecasting system using **XGBoost ML model with 86.5% accuracy**
- Delivered alerts in **12 Indian languages** enabling **95% faster response times**
- Reduced expected casualties by **70%** and used **Mesh technology** for SOS transfer
- Designed full-stack system using **FastAPI MongoDB React TypeScript and Docker**

[Enterprise Knowledge Assistant using RAG and Local LLMs](#)

- Built a **RAG-based knowledge assistant** using **Llama.cpp, ChromaDB, and Sentence Transformers**
- Implemented **semantic search, vector embeddings, and query rewriting** for context-aware retrieval
- Developed **incremental indexing and conversational memory** pipelines for multi-turn document QA

Education

Haridwar University

Sep 2023 – Jun 2027

Bachelor of Technology in Computer Science and Engineering **CGPA 8.00**

Indian Institute of Technology Mandi

Aug 2025 – Jun 2026

Minor in Artificial Intelligence and Data Science **CGPA: 8.70**

Certifications and Achievements

- Cleared CodeVita Round 2 with rank **270 among 100,000+** participants [Link](#)
- Reached **Top 25 National Finalists** at Trwizardthon Hackathon among **1,300+ teams and 5,200+ students from 28+ states** [Link](#)
- Reached **Top 10 out of 300+ teams** at AI Manthan 2025 **National Hackathon** hosted by IIIT Delhi [Link](#)
- Oracle Cloud Infrastructure 2025 Certified DevOps Professional [Link](#) and Data Science Professional [Link](#)